AttendSmart

Proxy-proof Offline Biometric Attendance

A mobile-first, offline-capable attendance system for colleges that prevents proxy attendance using face (smile) recognition, BLE proximity, and barometric floor checks. It's fast, secure, and easy for both students and teachers.

How It Works

Student Experience

- **Login:** Enter name, USN, DOB (no registration; teacher preloads data).
- **Check-In:**
    - **Face Detection:** Smile for the camera (liveness check).
    - **BLE Proximity:** Must be near teacher's device (within 8m).
    - **Barometer:** Confirms you're on the right floor.
- **Random Verification:** If flagged (e.g., too far), a random classmate is asked: "Is [Name] present? Yes/No". Their answer goes to the teacher for final approval.
- **Manual Fallback:** Teacher can manually mark attendance if needed (e.g., tech issues).
- **Feedback:** See your status instantly (present, absent, pending, manual).

Teacher Experience

- **Session Setup:** Schedule tomorrow's class (students see it in advance).
- **Start Check-In:** Open check-in window at class time.
- **Live Dashboard:** See real-time check-ins, flagged cases, and analytics.
- **Random Verification:** For flagged students, see peer's answer and approve/reject.
- **Manual Fallback:** Add attendance manually for special cases (syncs to student dashboard).

- **Reports:** Download attendance, spot trends, and get alerts for suspicious activity.

Key Technologies (and Why)

- **Flutter/Dart:** Fast, cross-platform mobile app.
- **Google ML Kit:** Face detection with smile liveness (prevents photo/video spoofing).
- **BLE (Bluetooth Low Energy):** Checks if student is physically near teacher.
- **Barometer:** Ensures student is on the right floor.
- **SQLite:** Secure, offline local storage (encrypted).
- **Node.js/Express + MongoDB:** Syncs non-sensitive data for analytics and dashboards.
- **React + Material-UI:** Real-time teacher/admin web dashboard.
- **QR Code (optional):** For large events/seminars only, not regular classes.

What Makes It Special

- **Proxy-proof:** Multi-factor (face+smile, BLE, barometer) + random peer check.
- **Offline-first:** Works without internet; syncs when online.
- **Privacy:** Biometric data never leaves device; only attendance status is shared.
- **Fast & Battery-friendly:** Check-in takes seconds, minimal power use.
- **Simple UI:** Clear, non-technical instructions for all users.
- **Manual Fallback:** Teacher can always override in special cases.

Future Add-ons (Ideas)

- Voice recognition check-in
- AI-powered engagement analytics
- Parent notifications
- Smart campus integrations (library, hostel, cafeteria)
- IoT device support

Team Takeaway

AttendSmart is a robust, easy-to-use, and secure attendance solution. It combines the best of biometrics, sensor tech, and human verification to make proxy attendance nearly impossible—while keeping things simple for everyone. Use this summary to understand the full flow, tech, and user experience at a glance.